Exploratory Data Analysis (EDA) Insights Report

BRFSS 2015 + 2021 Diabetes Health Indicators

Raw-dataset descriptive analysis

October 2026

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Analysis basis and reproducibility note

This report summarizes the combined, raw (unbalanced) survey dataset: **453,331 records and 25 original columns**, with no missing values or exact duplicate rows. The 2015 sample contains 229,781 records and the 2021 sample contains 223,550 records.

Income response options differ between the two survey years, so income is grouped onto a common scale for comparisons. The lower income brackets remain separate, while the highest brackets are combined into one **$75,000+** group. This includes respondents from both years and makes the upper-income category comparable.

All percentages are unweighted raw-sample percentages unless stated otherwise. Findings describe associations, not causation, and are not automatically nationally representative estimates.

1. Dataset overview and target distribution

The raw CSV contains **453,331 records and 25 columns**. The raw data check found **no missing values and no exact duplicate rows**. The two survey-year samples contain 229,781 records from 2015 and 223,550 records from 2021.

Plot 1. Distribution of diabetes status - `Diabetes_012`

Class	Meaning	Count	Percentage
0	No diabetes	374,597	82.63%
1	Prediabetes	10,242	2.26%
2	Diabetes	68,492	15.11%
Total		**453,331**	**100%**

Insight. Most respondents are coded as having no diabetes. Prediabetes is relatively uncommon (2.26%), while diabetes represents 15.11% of records. This shows why the three-class target is imbalanced, especially for prediabetes. Combining prediabetes and diabetes creates a larger positive class, but it also combines two medically different groups.

Plot 2. Binary target distribution - `Health_Status`

For the binary analysis, respondents with no diabetes form the No diabetes group. Respondents with prediabetes or diabetes are combined into the positive group, labelled At Risk.

Binary class	Count	Percentage
Healthy / target 0	374,597	82.63%
At Risk / target 1	78,734	17.37%
Total	**453,331**	**100%**

Insight. There are approximately 4.76 times as many negative examples as positive examples. This supports evaluating recall, precision, F1-score, PR-AUC and related class-sensitive metrics rather than relying on accuracy alone.

Terminology caution. "Healthy" means no diabetes according to `Diabetes_012`; it does not mean the person has no other health conditions. "At Risk" combines people coded as prediabetic and people coded as diabetic. The model therefore classifies a reported diabetes-related status, not whether someone will develop diabetes in the future.

Plot 3. Missing-value and duplicate checks

The raw CSV has no missing values and no exact duplicate rows according to pandas. Therefore, `dropna()` and `drop_duplicates()` should not change the row count for this version of the file. Keep the checks for reproducibility, but report their results rather than silently assuming rows were removed.

2. Single-variable distribution plots

These plots describe the composition of the raw dataset. Individually, they do **not** show whether a feature is associated with diabetes; that requires target-relationship plots.

Plot 4. High blood pressure - `HighBP`

- No high blood pressure: 252,419 respondents (55.68%).
- High blood pressure: 200,912 respondents (44.32%).

Insight. High blood pressure is common in the sample, affecting roughly 44% of respondents. It is a useful variable for target analysis, but the count plot alone cannot establish whether high blood pressure is more common among the positive target class.

Plot 5. High cholesterol - `HighChol`

- No high cholesterol: 259,956 respondents (57.34%).
- High cholesterol: 193,375 respondents (42.66%).

Insight. Approximately 43% of respondents report high cholesterol. Compare the target rate between the two groups to understand its descriptive association with diabetes-related status.

Plot 6. Cholesterol checks - `CholCheck`

- Had a cholesterol check: 435,427 respondents (96.05%).
- No cholesterol check: 17,904 respondents (3.95%).

Insight. Nearly all respondents are coded as having had a cholesterol check within the relevant period. The smaller group without a check may differ in age, healthcare access or existing diagnoses. This plot does not show that cholesterol checks cause or prevent diabetes.

Plot 7. BMI categories

BMI category	Count	Percentage
Underweight	5,834	1.29%
Normal	111,527	24.60%
Overweight	160,498	35.40%

BMI category	Count	Percentage
Obese	175,472	38.71%

Categories used: underweight <18.5; normal 18.5 to <25; overweight 25 to <30; obese >=30.

Insight. Overweight and obese respondents together account for approximately **74.11%** of records. Obesity is the largest individual BMI group. The categories make the pattern easier to explain but discard some detail compared with analysing BMI numerically.

Plot 8. BMI histogram

- Mean BMI: 28.91.
- Median BMI: 28.
- 25th percentile: 24.
- 75th percentile: 32.
- Minimum: 12.
- Maximum: 99.

Insight. Central BMI values are above the usual normal-weight range, and the mean is slightly higher than the median, reflecting some upper-tail skew. Only 1,352 respondents (0.30%) have BMI >=60, and 325 (0.07%) have BMI >=80. The maximum of 99 deserves a data-quality check, but extreme values should not be deleted automatically without checking source coding. This is a count histogram with a KDE overlay; the y-axis represents respondent count.

Plot 9. General health - `GenHlth`

General health	Count	Percentage
Excellent	71,094	15.68%
Very Good	156,333	34.49%
Good	145,687	32.14%
Fair	59,350	13.09%
Poor	20,867	4.60%

Insight. "Very Good" and "Good" together account for 66.63% of the sample. Approximately 17.69% report Fair or Poor general health. The later target analysis shows a strong gradient in the positive-target rate as self-reported health worsens.

Plot 10. Physical-health days - `PhysHlth`

Physical-health days	Count	Percentage
0 days	280,983	61.98%
1-5 days	87,004	19.19%
6-14 days	28,960	6.39%
15-30 days	56,384	12.44%

Insight. Most respondents report no physically unhealthy days, but 12.44% report 15-30 days. Compare these categories against the target to see whether physical-health burden corresponds to different observed positive-target rates.

Plot 11. Mental-health days - `MentHlth`

Mental-health days	Count	Percentage
0 days	287,672	63.46%
1-5 days	85,779	18.92%
6-14 days	29,512	6.51%
15-30 days	50,368	11.11%

Insight. Most respondents report no mentally unhealthy days, while 11.11% report 15-30 days. The distribution is broadly similar to the physical-health distribution, but the two measures are not interchangeable. The target relationship for mental-health days is less straightforward than the physical-health pattern.

Plot 12. Smoking status - `Smoker`

- Not coded as a smoker: 251,121 (55.39%).
- Coded as a smoker: 202,210 (44.61%).

Insight. Smoking is common enough to warrant further analysis. The binary indicator is simplified and should not be interpreted as distinguishing current, former and never smokers unless those categories are separately represented in the dataset.

Plot 13. Stroke history - `Stroke`

- No stroke history: 433,855 (95.70%).
- Stroke history: 19,476 (4.30%).

Insight. Stroke history is uncommon compared with the other class. A count chart can make it look visually small, so a comparison of positive-target percentages is more useful for understanding its association.

Plot 14. Heart disease or heart attack - `HeartDiseaseorAttack`

- No: 409,253 (90.28%).
- Yes: 44,078 (9.72%).

Insight. About one in ten respondents is coded as having heart disease or a heart attack. Later target analysis shows a considerably higher positive-target rate for respondents in this group, although the association is observational.

Plot 15. Physical activity - `PhysActivity`

- Physical activity: 340,004 (75.00%).
- No physical activity: 113,327 (25.00%).

Insight. Three quarters of the sample are coded as physically active. The target-rate plot can compare the two groups, but cannot establish a causal effect of activity.

Plot 16. Fruit consumption - `Fruits`

- Yes: 276,743 (61.05%).
- No: 176,588 (38.95%).

Insight. Most respondents are coded as consuming fruit according to this binary indicator. It does not capture portion sizes or the full quality of an individual's diet.

Plot 17. Vegetable consumption - `Veggies`

- Yes: 365,716 (80.67%).
- No: 87,615 (19.33%).

Insight. Vegetable consumption is more prevalent than fruit consumption according to the recorded binary indicators. The plot does not show whether respondents meet a particular dietary recommendation.

Plot 18. Heavy alcohol consumption - `HvyAlcoholConsump`

- Not coded as heavy alcohol consumption: 424,793 (93.70%).
- Coded as heavy alcohol consumption: 28,538 (6.30%).

Insight. The positive group is relatively uncommon. Later analysis shows a lower crude positive-target rate among people coded in this group. Do **not** interpret this counterintuitive association as evidence that heavy alcohol consumption protects against diabetes.

Plot 19. Healthcare coverage - `AnyHealthcare`

- Healthcare coverage: 432,105 (95.32%).
- No healthcare coverage: 21,226 (4.68%).

Insight. Most respondents report healthcare coverage. The smaller uncovered group may produce less stable estimates and may differ in age, health needs or healthcare access.

Plot 20. Could not see a doctor due to cost - `NoDocbcCost`

- No: 416,941 (91.97%).
- Yes: 36,390 (8.03%).

Insight. About 8% report cost-related difficulty seeing a doctor. This indicator describes access barriers; its target association is not necessarily causal.

Plot 21. Walking difficulty - `DiffWalk`

- No difficulty: 374,349 (82.58%).
- Difficulty: 78,982 (17.42%).

Insight. Approximately one in six respondents reports walking difficulty. This feature shows one of the largest differences in the later target analysis. Age and other health conditions may also influence the association.

Plot 22. Sex - `Sex`

Using the current notebook mapping:

- Female / code 0: 246,005 (54.27%).
- Male / code 1: 207,326 (45.73%).

Insight. The groups are reasonably balanced. The target-rate comparison shows a modest difference between them, which should not be treated as a causal or complete biological explanation.

Plot 23. Age categories - `Age`

The age variable contains 13 age groups, from 18-24 through 80+. The largest groups are age 60-64 with 55,365 respondents (12.21%) and age 65-69 with 55,325 respondents (12.20%). The smallest group is age 18-24, with 12,164 respondents (2.68%).

Insight. The sample contains more respondents in older age groups than in the youngest group. Overall diabetes-related prevalence depends partly on this sample age composition.

Plot 24. Education - `Education`

Education category	Count	Percentage
Never attended / kindergarten only	321	0.07%
Grades 1-8	6,809	1.50%
Some high school	16,405	3.62%
High school graduate	113,136	24.96%
Some college	130,191	28.72%
College graduate	186,469	41.13%

Education category	Count	Percentage

Insight. College graduates are the largest education group, accounting for 41.13% of respondents. Some lower education groups are small; estimates for them should be interpreted with greater caution.

Plot 25. Income distribution - `Income`

Why harmonization matters. Income response options differ between the two survey years. The highest income brackets are grouped together so that respondents from both years can be included in a comparable upper-income category.

The highest income brackets are combined into one shared `$75k+` group. The income categories have the following raw counts and unadjusted At-Risk rates:

Harmonized income category	Count	Share of sample	At-Risk rate
< $10k	15,655	3.45%	27.66%
$10k-$15k	18,631	4.11%	30.08%
$15k-$20k	24,761	5.46%	25.63%
$20k-$25k	32,607	7.19%	23.64%
$25k-$35k	52,148	11.50%	21.09%
$35k-$50k	66,258	14.62%	18.23%
$50k-$75k	79,931	17.63%	15.77%
$75k+	163,340	36.03%	11.67%

Insight. In the pooled sample, the observed At-Risk rate is generally lower at higher harmonized income levels, with a small non-monotonicity in the first two groups. These are unadjusted associations and should not be interpreted as a causal income effect. The same income grouping should be used throughout the income-related analyses for consistent comparisons.

Official references: CDC BRFSS 2015 codebook and CDC BRFSS 2021 codebook.

3. Relationship plots

These plots compare diabetes status with other variables. Percentages in the tables below are within each feature group.

Plot 26. Correlation heatmap of health indicators

Largest recorded correlations with `Diabetes_012` include:

Feature	Correlation with `Diabetes_012`
General health (`GenHlth`)	+0.282
High blood pressure (`HighBP`)	+0.263
Walking difficulty (`DiffWalk`)	+0.210
BMI	+0.208
High cholesterol (`HighChol`)	+0.204
Age	+0.193
Heart disease or attack	+0.172
Physical-health days (`PhysHlth`)	+0.158
Harmonized income (`Income_Harmonized`)	-0.144
Physical activity (`PhysActivity`)	-0.122
Education	-0.102

Insight. The most visible linear associations are with general health, high blood pressure, walking difficulty, BMI, high cholesterol and age. Harmonized income, physical activity and education have negative correlations under their current numeric/ordinal coding. Correlation measures association, not causation; Pearson correlation on ordinal codes is only a descriptive approximation.

Important issue. The heatmap includes both `Diabetes_012` and `Health_Status`, even though the latter is constructed directly from the former. Their correlation is approximately 0.981 because they encode nearly the same underlying outcome. This is a target-mapping effect, not a newly discovered medical relationship. Exclude both target columns from a predictor-only correlation matrix and never use them as model inputs to predict `Health_Status`.

Plot 27. BMI boxplot by diabetes class

`Diabetes_012` class	Mean BMI	Median BMI	Middle 50% of BMI values
No diabetes	28.26	27	24-31
Prediabetes	31.14	30	26-35
Diabetes	32.08	31	27-36

Insight. BMI rises across the three groups in the observed sample. Median BMI increases from 27 in the no-diabetes group to 30 for prediabetes and 31 for diabetes. The distributions overlap, so BMI alone cannot perfectly distinguish status.

Plot 28. Diabetes prevalence by high blood pressure

Blood-pressure group	No diabetes	Prediabetes	Diabetes
No high blood pressure	91.43%	1.61%	6.97%
High blood pressure	71.58%	3.08%	25.34%

Insight. Diabetes is recorded for 25.34% of respondents with high blood pressure, compared with 6.97% without high blood pressure. The binary positive-target rates are 28.42% versus 8.57%. This is a clear descriptive association, but it does not prove that high blood pressure causes diabetes.

Plot 29. Diabetes prevalence by high cholesterol

Cholesterol group	No diabetes	Prediabetes	Diabetes
No high cholesterol	89.32%	1.58%	9.10%
High cholesterol	73.64%	3.17%	23.18%

Insight. The diabetes rate is 23.18% among respondents with high cholesterol versus 9.10% among those without. The combined positive-target rates are 26.36% and 10.68%, respectively. This remains observational.

Plot 30. Diabetes prevalence by BMI category

BMI category	No diabetes	Prediabetes	Diabetes	Combined positive target
Underweight	93.20%	1.03%	5.78%	6.80%
Normal	91.97%	1.30%	6.73%	8.03%
Overweight	85.42%	2.01%	12.58%	14.58%
Obese	73.80%	3.14%	23.06%	26.20%

Insight. The combined positive-target rate rises from 8.03% among normal-BMI respondents to 26.20% among respondents in the obese category. Prediabetes percentage also rises from 1.30% to 3.14%. This agrees with the BMI boxplot while showing target composition rather than BMI distributions.

Plot 31. Diabetes prevalence by walking difficulty

Walking difficulty	No diabetes	Prediabetes	Diabetes
No difficulty	86.23%	2.02%	11.75%
Difficulty	65.57%	3.39%	31.04%

Insight. Respondents with walking difficulty have a 31.04% diabetes rate, compared with 11.75% among those without it. The combined positive-target rates are 34.43% and 13.77%. Age, general health, physical-health burden and BMI may contribute to this association.

Plot 32. Heart disease by age and income

Heart-disease prevalence rises sharply across age groups in the raw sample: approximately 0.51% for ages 18-24, 10.66% for ages 60-64, 16.64% for ages 70-74 and 24.00% for age 80+. Using the harmonized income field, lower-income groups generally have higher observed heart-disease rates than higher-income groups. Overall rates range from about 15-18% in the lowest categories to around 6% in the `$75k+` category.

Insight. Age is strongly associated with reported heart disease. These income-specific values use the shared income grouping across both survey years. They are unadjusted sample comparisons.

Plot 33. Diabetes status by age and income

The mean of the numeric three-class outcome `Diabetes_012` is not a direct diabetes prevalence or At-Risk proportion, because its classes represent different categories. The values reported here use `Health_Status` for the combined binary target, rather than treating the mean of `Diabetes_012` as prevalence. The positive-target rate rises from 2.12% at ages 18-24 to 26.08% at ages 70-74, then declines to 22.31% for age 80+. With harmonized income, lower-income groups generally show higher rates within many age groups. These are crude, unadjusted comparisons.

Plot 34. Mutual-information feature ranking

This ranking uses `Diabetes_012` as the target and therefore describes information about the three-class outcome, not the binary `Health_Status` target. It is the notebook's original three-class baseline and uses the original `Income` column rather than `Income_Harmonized`. Because income response options differ between survey years, its income-related score is not directly comparable with Plot 43's harmonized-income score. Feature-type settings also matter when categorical or ordinal variables are stored as numeric values.

Insight. Mutual information is a useful exploratory ranking method. For the binary target, Plot 43 reports a separate ranking using `Health_Status` and explicit feature-type handling. Scores are estimates, not causal importance or percentages of model performance.

4. Target-focused EDA plots

Plot 35. BMI versus binary Health Status - straight-line XY plot

The plot places BMI on the x-axis and `Health_Status` on the y-axis, with a straight trend line. Because the target has only values 0 and 1, points form two horizontal bands. For readability, the displayed scatter uses a reproducible sample of 15,000 records (`random_state=42`), or the full dataset if it contains fewer records. The trend line is fitted on that sample.

Insight. The fitted line is only a descriptive linear trend; it is not a smooth individual-level response and does not prove that BMI causes diabetes-related status. The next plot is generally easier to interpret.

Plot 36. Observed At-Risk percentage by BMI

BMI	Respondent count	At-Risk percentage
20	10,420	5.95%

BMI	Respondent count	At-Risk percentage
25	28,346	11.66%
30	26,844	19.56%
35	11,355	27.50%
40	4,892	33.12%
45	1,790	38.60%

Insight. The observed positive-target percentage generally rises with BMI. The increase is not perfectly linear, and the sample sizes shrink at higher BMI values. The plotted points include BMI values with at least 30 respondents, and the straight-line trend is weighted by the square root of respondent count. The line describes a broad trend rather than an exact rule for individual predictions.

Plot 37. At-Risk percentage by binary health indicators

Health indicator	Rate at value 0	Rate at value 1
High blood pressure	8.57%	28.42%
High cholesterol	10.68%	26.36%
Cholesterol check	3.02%	17.96%
Smoker	15.78%	19.34%
Stroke	16.59%	34.64%
Heart disease or attack	15.29%	36.64%
Physical activity	25.22% when no activity	14.75% when active
Fruit consumption	18.44% when no	16.69% when yes
Vegetable consumption	20.53% when no	16.61% when yes
Heavy alcohol consumption	18.01% when no	7.86% when yes
Healthcare coverage	12.92% when no	17.59% when yes
Could not see doctor due to cost	17.15% when no	19.88% when yes
Walking difficulty	13.77%	34.43%
Sex, using current code mapping	16.24% for code 0	18.70% for code 1

Insight. The largest crude differences appear for heart disease or attack, stroke, walking difficulty, high blood pressure and high cholesterol. Physical activity and fruit/vegetable indicators have lower crude positive-target rates among respondents coded as reporting those behaviors.

Some crude findings are counterintuitive, including lower positive-target prevalence in the heavy-alcohol group and higher prevalence among people reporting healthcare coverage. Do not describe these as protective or harmful effects. Age, diagnosis patterns, healthcare engagement, selection effects and other factors may affect these unadjusted rates.

Plot 38. At-Risk percentage across age groups

Age group	At-Risk percentage
18-24	2.12%
25-29	3.25%
30-34	4.05%
35-39	6.17%
40-44	8.85%
45-49	12.26%
50-54	15.61%
55-59	18.61%
60-64	21.77%
65-69	24.20%

Age group	At-Risk percentage
70-74	26.08%
75-79	25.85%
80+	22.31%

Insight. The positive-target rate rises from about 2% in the youngest group to approximately 26% among ages 70-74, then declines in older groups. The decline should not be interpreted as age becoming protective; it may reflect sample composition, survival, reporting or other factors. This is a cross-sectional survey, not a record of individuals ageing over time.

Plot 39. At-Risk percentage by self-reported general health

General health	At-Risk percentage
Excellent	4.48%
Very Good	10.10%
Good	20.88%
Fair	34.89%
Poor	41.31%

Insight. This is one of the clearest gradients: the positive-target rate rises consistently as reported general health worsens, from 4.48% for Excellent to 41.31% for Poor. Diabetes and related comorbidities may contribute to poorer self-rated health, but other health conditions and respondent characteristics matter. The chart does not establish direction or causality.

Plot 40. Health-status composition by survey year

Target category	2015	2021
No diabetes	82.71%	82.55%
Prediabetes	2.01%	2.51%
Diabetes	15.27%	14.94%
Binary positive target	**17.29%**	**17.45%**

Insight. The binary positive-target proportions are similar across the two samples, differing by about 0.16 percentage points. The prediabetes share is higher in the 2021 sample (2.51% versus 2.01%), while the diabetes share is slightly lower (14.94% versus 15.27%). This compares two survey-year samples, not a continuous trend. These raw proportions are not automatically official population estimates; survey weights and design need to be considered for population-level inference.

Plot 41. At-Risk percentage by age and income

Insight. After harmonizing income codes, positive-target percentages generally increase with age and tend to be higher in lower-income categories within many age groups. For respondents aged 60-64, the recorded rate is 35.5% in the lowest income category versus 16.1% in the harmonized `$75k+` category. Among ages 70-74, the rates are 37.7% and 20.2%, respectively.

This suggests that age and income jointly describe differences in the observed target rate, but the heatmap is unadjusted. Some age-income cells may be less stable than others. Harmonize income before drawing conclusions from this plot.

Plot 42. Combined health profile - HighBP, HighChol and DiffWalk

HighBP	HighChol	DiffWalk	Count	At-Risk percentage
1	1	1	34,280	47.19%
1	0	1	18,465	30.87%
1	1	0	82,582	29.17%
0	1	1	10,530	26.69%
1	0	0	65,585	16.97%
0	0	1	15,707	15.95%
0	1	0	65,983	11.96%
0	0	0	160,199	5.26%

Here, 1 means Yes and 0 means No. Only groups with at least 100 respondents are included.

Insight. Respondents with all three indicators have the highest positive-target rate (47.19%), while the group with none has the lowest (5.26%). Combinations reveal patterns that single-variable plots can miss. This does not prove that the three factors independently cause diabetes; age, BMI and other characteristics may influence the combinations.

Plot 43. Mutual information with `Health_Status`

The calculation uses the binary target, a reproducible sample of 100,000 records (`random_state=42`), and the harmonized income field. The estimation treats BMI, mental-health days (`MentHlth`), and physical-health days (`PhysHlth`) as continuous features. The remaining coded indicator/ordinal features are treated as discrete. Approximate leading scores are:

Rank	Feature	Mutual-information score
1	General health	0.0395
2	High blood pressure	0.0337
3	Age	0.0235
4	BMI	0.0224
5	High cholesterol	0.0205
6	Walking difficulty	0.0191
7	Heart disease or attack	0.0118
8	Harmonized income	0.0104
9	Physical-health days	0.0097
10	Physical activity	0.0066

Ranking note: when sorted strictly by the displayed score, harmonized income (0.0104) ranks above physical-health days (0.0097), so the order is: heart disease or attack, harmonized income, physical-health days, then physical activity. Mutual information is an exploratory estimate, not a percentage of model performance and not causal importance. Scores can vary with sample and estimation settings.

Plot 44. Predictor-only correlation heatmap

The predictor-only correlation heatmap excludes `Diabetes_012` and `Health_Status` so that target columns are not included in the predictor matrix. Recorded examples include high blood pressure with high cholesterol (approximately $r = 0.280$) and high blood pressure with BMI (approximately $r = 0.197$).

Insight. Some predictors are related to each other and may carry overlapping information. Correlated predictors are not automatically problematic for an ANN, but the heatmap helps describe the data and compare relationships with target-focused rankings. A correlation heatmap is not a causal diagram.

5. What the plots tell us together

1. **The binary target is substantially imbalanced.** Only 17.37% of the raw sample is in the combined prediabetes/diabetes class. This supports comparing imbalance-handling strategies, while evaluating positive-class performance carefully.

2. **Several health indicators show clear crude associations with target status.** General health, high blood pressure, BMI, walking difficulty, high cholesterol, age and heart disease or attack repeatedly appear among the strongest descriptive relationships.
3. **The age pattern is strong but not perfectly monotonic.** The positive-target rate rises through ages 70-74 and then declines among older groups. Report this as a cross-sectional association, not the effect of ageing on an individual.
4. **Income and education show socioeconomic differences, but the income comparisons require a common grouping across the two survey years.** With a common income scale, the raw sample generally shows lower positive-target rates at higher income and education levels. These are unadjusted associations and may reflect age, access to care and other factors.
5. **Lifestyle indicators show nuanced patterns.** Physical activity and fruit/vegetable indicators are associated with lower crude rates, while heavy alcohol consumption and healthcare coverage show counterintuitive crude results. These must not be interpreted as protective or harmful causal effects.
6. **No single variable perfectly separates the target classes.** BMI and high blood pressure are informative, but the distributions and target rates overlap. Combining multiple demographic, behavioral and health features is reasonable for ANN modeling.

6. Reproducibility and reporting checks

1. **Use a consistent income grouping for cross-year comparisons.** Combine the highest income brackets into the shared `$75k+` category in the income distribution, age-income heatmap, socioeconomic comparisons, predictor correlation, and Plot 43's binary-target mutual-information analysis. Plot 34 is retained as the notebook's original three-class baseline using raw `Income`; its income-related score is therefore not directly comparable with Plot 43. For a fully harmonized three-class ranking, use `Income_Harmonized` in that baseline as well.
2. **Use the appropriate target for target-rate plots.** For the combined positive class, calculate the mean of `Health_Status`. The mean of `Diabetes_012` is not a prevalence measure. For diabetes-only prevalence, use a separate indicator such as `Diabetes_012 == 2`.
3. **Exclude target-derived fields from predictor analyses.** Neither `Diabetes_012`, `Health_Status` nor a label derived from `Health_Status` should be used as an ANN input when predicting `Health_Status`. The predictor-only heatmap should exclude target columns.
4. **Report the original dataset size accurately.** The source CSV has 25 original columns. The notebook's home-page metric currently reads `df.shape[1]` after helper and label columns have been added, so it can report more than the original 25 columns. Store the original column count immediately after loading the CSV, or calculate the model-input count from an explicit predictor list.
5. **Use precise plot labels.** The BMI histogram displays respondent counts, and age-specific percentages should be described as raw-sample rates rather than population percentages.
6. **Keep the analysis data and definitions consistent.** Use the same raw dataset and harmonized income definition across related plots and calculations so that the reported results can be reproduced.
7. **Avoid causal and population-level claims from unweighted plots.** These charts show descriptive associations in the combined survey sample. Survey weights, design variables and appropriate statistical analysis are needed for population inference. An observational EDA alone does not identify causal effects.

Final conclusion

The full raw dataset is imbalanced, and diabetes-related status varies substantially across several health indicators. The clearest descriptive gradients involve general health, age, BMI, high blood pressure, high cholesterol, walking difficulty and heart disease. The two included survey-year samples have similar raw binary-target percentages. With a common income grouping, positive-target rates are generally lower in higher-income groups, although these are unadjusted associations. The findings support investigating multiple predictors in ANN models, but they do not establish causal relationships or show that any variable alone can diagnose diabetes. Income-related comparisons use the same shared grouping across both survey years, and all raw proportions should be interpreted as sample descriptions rather than automatically population-representative estimates.